

# **Enhancing Multimedia Forensics and Authentication Analysis through Enhanced Explainability and Summarization of Results**

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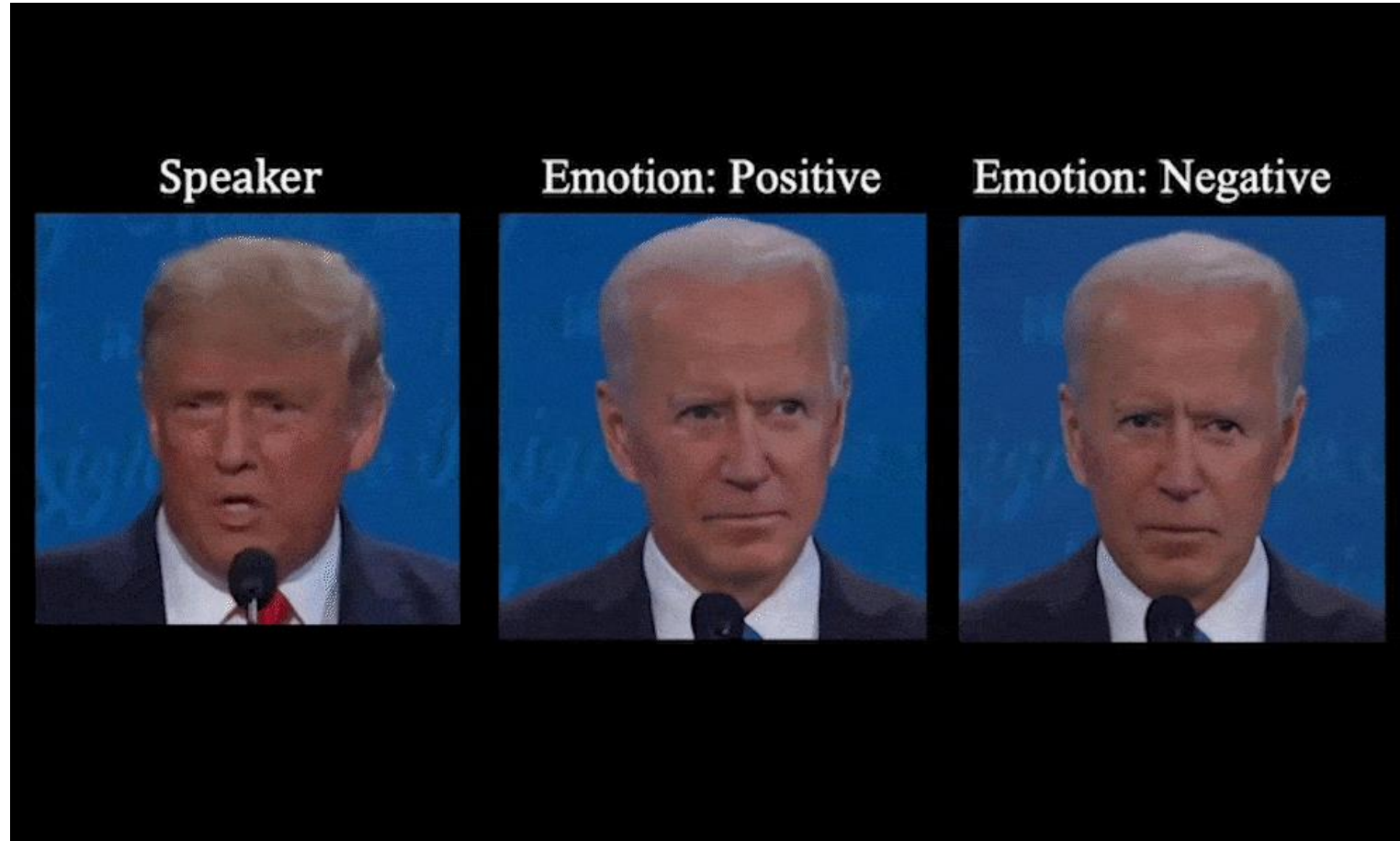
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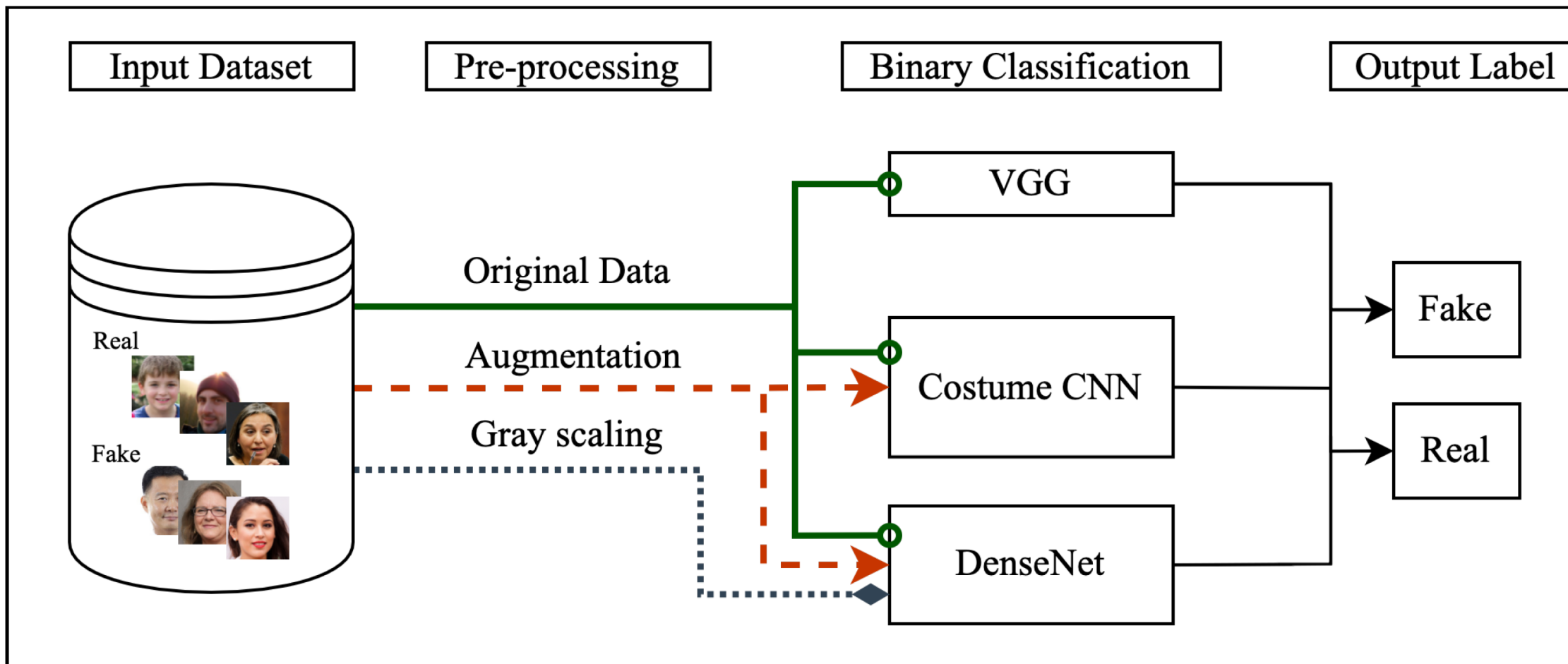
# **Part 1 Introduction of Explainable Forensics**

# Video Manipulation Detection is crucial



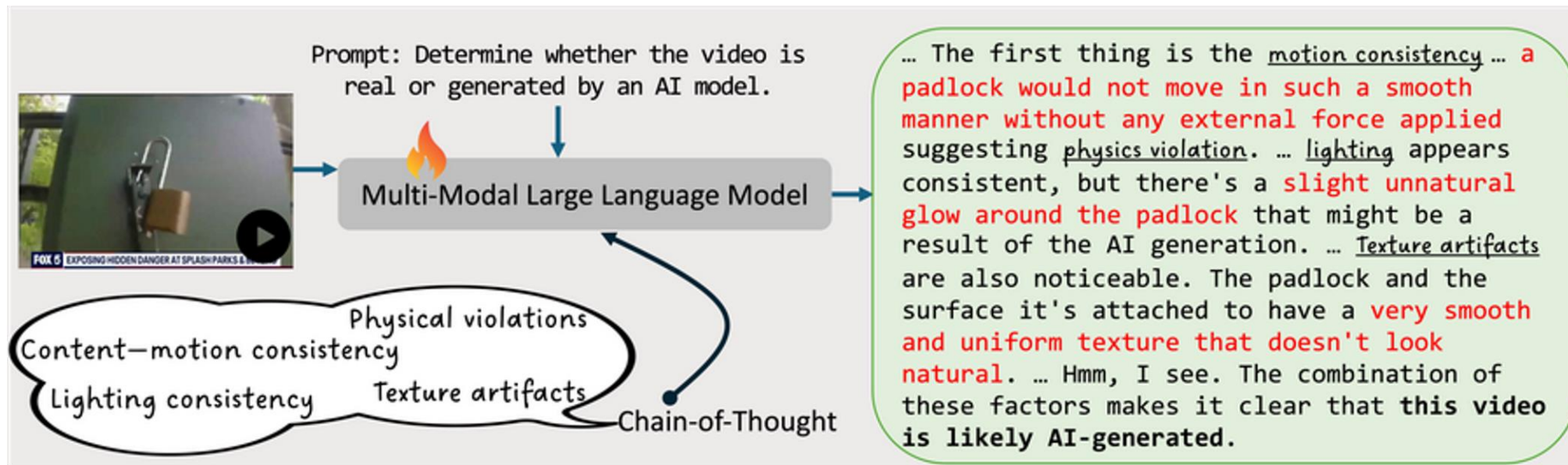
- Altered videos can be extremely dangerous

# Current detection tools lack explanation



# What is Explainable Forensics AI

- AI analysis for generating explanations



# **Part 2 General-manipulation Detection Dataset**

# From face-manipulated to general-manipulated

## ExDDV



There are small left-right jitters of the eyes. The mouth shows distortions.



There is distortion of all facial features caused by overlapping artifacts. The eyebrows are darting up and down. A residual set of eyebrows is noticeable. Residuals from the source face flash across at times causing a change in the look of the features.



The mouth is blurred of a few moments. The details of the face are not sharp.



This video is visually coherent.

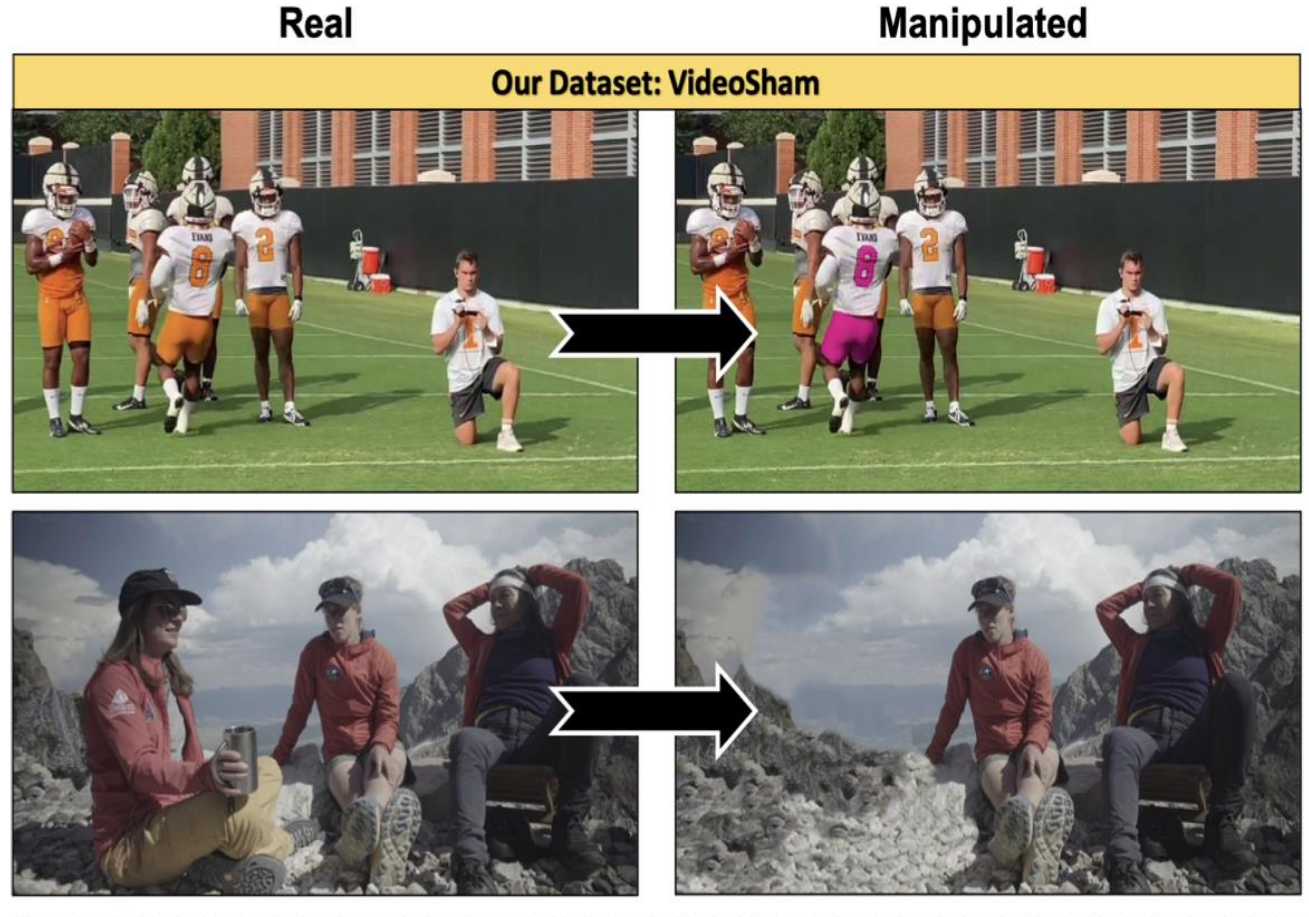
## VideoSham (w. Explanation)



Explanation: The video shows signs of digital manipulation, with a subject on the left side of the frame likely removed, evidenced by spatial inconsistencies, blurring artifacts, and a missing interaction dynamic.

# Selected Dataset: VideoSham

- Adobe Research Team
- Beyond Faces
- Total 6 manipulations



# Manipulation I: Adding an entity/subject

Real



Manipulated



## Original Description:

Around 0:42, there is a person in a black coat who passes the last turnstile, can you cut this person out, and make them cross the turnstile at a different time and different turnstile in the video

## Enriched Explanation:

The edit creates artifacts and spatial inconsistencies with a subject twice in the scene.

# Manipulation 2: Removing an entity/subject

Real



Manipulated



## Original Description:

Remove the man on the left side (00:00 - 00:05)

## Enriched Explanation:

The video shows signs of digital manipulation, with a subject on the left side of the frame likely removed, evidenced by spatial inconsistencies, blurring artifacts, and a missing interaction dynamic.

# Manipulation 3: Background/Color Change

**Real**



**Manipulated**



## Original Description:

Change the color of the cabinets behind the man

## Enriched Explanation:

The video appears to have been manipulated, with the color of the cabinets in the background possibly digitally altered, leading to color mismatches and flickering.

# Manipulation 4: Text Replaced/Added

Real



Manipulated



## Original Description:

0:00 to 0:06 change the text in the blue box to lorem ipsum

## Enriched Explanation:

The text in the blue box at the bottom of the screen was replaced with lorem ipsum, showing subtle blending artifacts, suggesting a text replacement.

# Manipulation 5: Inter-Frame Manipulations

Real



Manipulated



## Original Description:

Duplicate random frames in between to make it into a 10 second video (00:10 - 00:16)

## Enriched Explanation:

The video shows possible temporal manipulation, exhibited by stuttering motions, jerky movements and duplicated sequences which suggest manipulation techniques of frame duplication.

# Manipulation 6: Audio Replaced/Added

**Real**



**Manipulated**



## Original Description:

Trim to (3:05 - 3:10) and remove the guitar noise. Place the noise of a cow moo-ing instead.

## Enriched Explanation:

The audio track has been replaced, introducing the sound of a cow mooing which is inconsistent with the visual action of a guitar being played.

# Prompt-Engineering for Better Annotation

**Version 1:** The primary manipulation is the removal of the man on the left. The most evident visual anomaly is the lack of interaction with the man on the right, which highlights the missing participant. Slight inconsistencies in texture, sharpness, or blurring may appear along the boundary of the area where the man was removed, but the rest of the scene appears unaffected.

**Version 2:** The man on the left was digitally removed, causing background reconstruction artifacts, edge blurring, and a noticeable interaction anomaly with the remaining man.

**Version 3:** The video appears manipulated, with the man on the left side of the frame digitally removed, leading to missing interactions and spatial inconsistencies.

**Version 4:** The video shows signs of digital manipulation, with a subject on the left side of the frame likely removed, evidenced by spatial inconsistencies, blurring artifacts, and a missing interaction dynamic.

# **Part 3 Towards Better Detection Model**

# Evaluation on Qwen3 Omni



Prompt: Detect whether this video is manipulated? If yes, tell the manipulated location [x\_min, y\_min, x\_max, y\_max] and give me the reason for the detection.

# Evaluation on Qwen3 Omni

**No, this video is not manipulated.** The scene is a natural outdoor setting with a person standing in front of a building. The lighting, shadows, and details such as the clothing, hair, and background elements like the notice board and window appear consistent and realistic, with no signs of digital alterations or inconsistencies that would suggest manipulation.

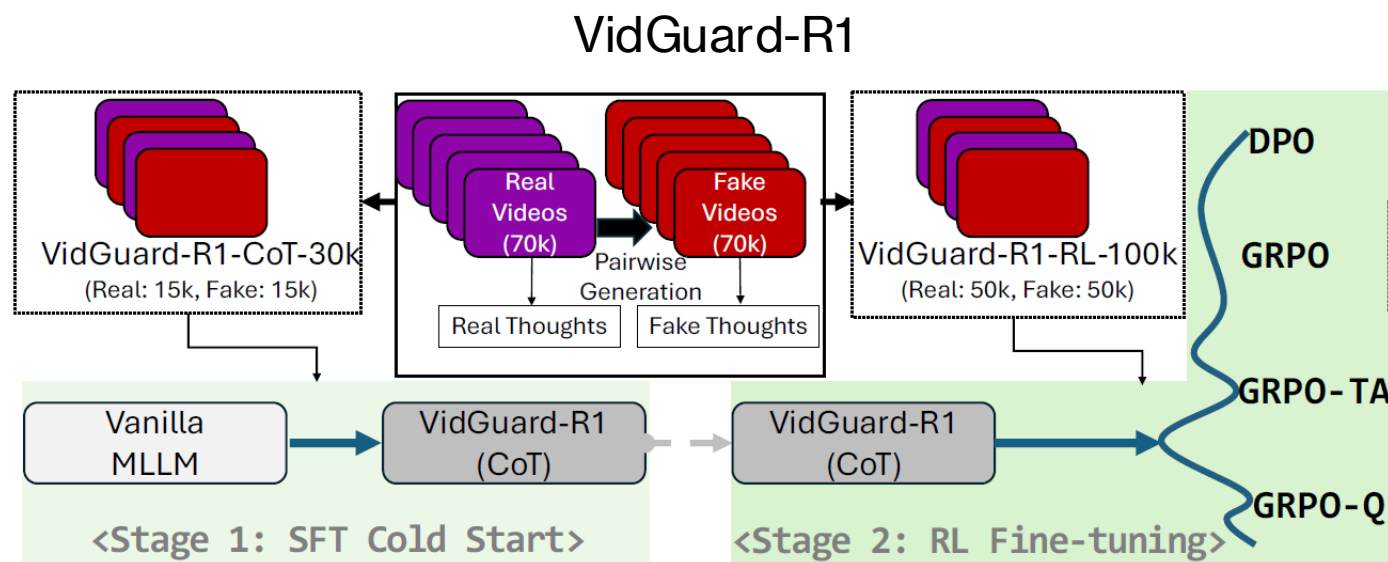
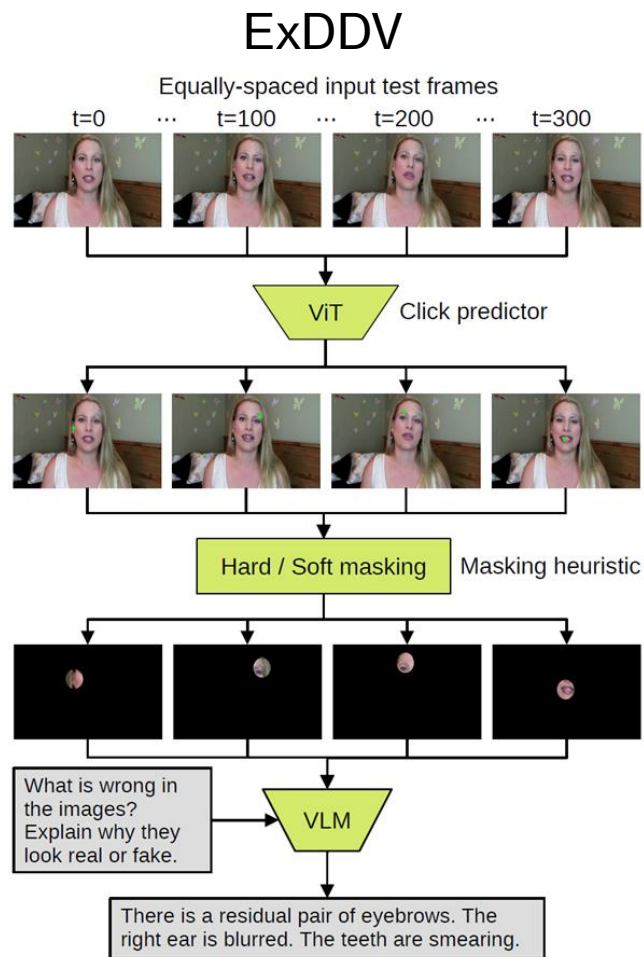
Qwen3 Omni Prediction **X**

The video appears manipulated; the individual on the left side of the frame has been removed between 00:00 and 00:05, leading to spatial inconsistencies and unnatural gaps in the scene.

Ground Truth Annotation

- Qwen3 Omni is not effective for forensics detection.

# Limitations on others: not support vision-audio



- We aim to build an effective model that support omni-manipulation detection.

# Project Deliverables and Challenges

- Completed
  - Project Scope and Dataset Review 9/15
  - Dataset Enrichment 10/13
  - Midterm Presentation 10/30
- In Progress
  - Detection Prototype Establishment 11/10
  - Retrieval-Augmented Explainability System 11/17
  - Final Report 11/28
- Challenges
  - Labor work on screening annotations
  - Current MLLM models are too weak

**Thank you!**

**Questions are welcome!**